Topic 13: What Regression Lines Do (OLS in Practice)

Argument, Data, and Politics - POLS 3312

Tom Hanna

2026-04-12

Table of contents

1. Agenda and recap

1.1 Today

- What a regression line (OLS) does, in pictures
- How the line becomes a prediction formula
- How to read a simple journal-article regression table
- How to turn coefficients into a prediction for a change in (X)
- Brief: what if the model is not OLS? (logit, probit, etc.)

1.2 Where we are in the course

- We have talked about variables, distributions, and relationships.
- We have seen scatterplots and talked about "positive" and "negative" relationships.
- Today we move from describing a relationship to writing *a specific line* that summarizes it.

2. What OLS is doing (no heavy math)

2.1 A cloud of points and a line

- Imagine a scatterplot of (X) (e.g., income) and (Y) (e.g., turnout).
- Many lines could be drawn, but some will "fit" the points better than others.
- Ordinary Least Squares picks **one** line as "best."

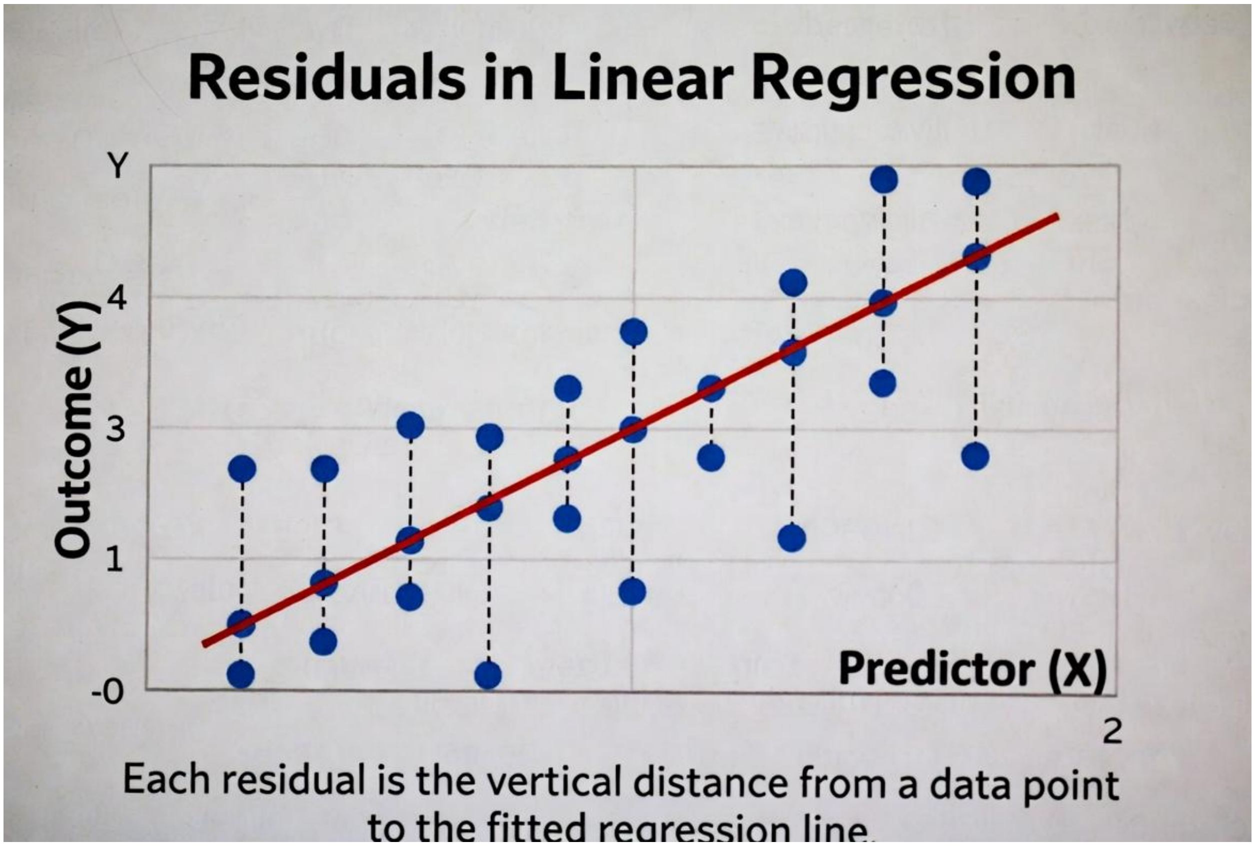
2.2 "Best" in the OLS sense

- For each point, the *residual* is "actual (Y) minus predicted (Y)."
- OLS chooses the line that makes the *squared residuals* as small as possible overall.
- Visually: the chosen line keeps the vertical distances from the points to the line as small as possible *on average*.

2.3 Video example

Show the video example here

2.4 Visual example (graphic)



Regression line drawn through a cloud of points, showing vertical distances (residuals) from each point to the line

3. Turning the line into a formula

3.1 The basic regression equation

- The line can be written as a formula:
  [= a + bX]
- (a) is the **intercept** (predicted (Y) when (X = 0)).
- (b) is the **slope** (how much we expect (Y) to change when (X) increases by 1 unit).

3.2 Interpreting the slope

- If (b = 2), then when (X) increases by 1 unit, predicted (Y) increases by 2 units.
- If (b = -0.5), a 1-unit increase in (X) is associated with a 0.5-unit *decrease* in predicted (Y).

- The slope is the “effect” of (X) on (Y) in this simple model.

3.3 Example with a story

- Suppose (Y =) political knowledge score (0–10) and (X =) years of education.
- If the estimated line is (= 2 + 0.3X), then:
  - A person with 10 years of education has predicted knowledge (= 2 + 0.3(10) = 5).
  - A person with 14 years of education has predicted knowledge (= 2 + 0.3(14) = 6.2).
- Going from 10 to 14 years (a 4-year increase) raises predicted knowledge by (0.3 = 1.2) points.

4. Reading a simple regression table

4.1 What a one-variable regression table looks like

In a journal article, a very simple OLS table for one (X) might look like this:

Variable	Coefficient	Std. Error
Education (years)	0.30	0.05
Constant	2.00	0.40

- “Coefficient” is the estimate of (b) (for Education) or (a) (for the Constant).
- “Std. Error” is a measure of uncertainty, but today we focus on using the **coefficients** for predictions.

4.2 Turning the table into the formula

- From the table, the **intercept** is 2.00.
- The **slope** for Education is 0.30.
- So the regression line is: [= 2.00 + 0.30]

4.3 Using the formula for prediction

- If a person has 12 years of education:
  - (= 2.00 + 0.30(12) = 5.6).
- If a person has 16 years of education:
  - (= 2.00 + 0.30(16) = 6.8).
- The model gives a *predicted* knowledge score for any value of education.

5. From change in X to change in Y

5.1 The key idea

- In a simple line (= a + bX), the slope (b) tells us the change in predicted (Y) for a 1-unit change in (X).
- For a bigger change in (X), multiply (b) by the size of that change.
- This is often what authors mean by "substantive effect."

5.2 Step-by-step method students should use

1. Read the coefficient for the variable (X) of interest (this is (b)).
2. Decide on a realistic change in (X) (for example, 4 extra years of education).
3. Multiply the coefficient by that change in (X).
4. Interpret the result as the predicted change in (Y).

5.3 Worked example

Using the table above:

- Coefficient on Education: (b = 0.30).
- Consider a change in Education from 10 to 14 years ((X = 4)).
- Predicted change in (Y) is: [= b X = 0.30 = 1.2]
- Interpretation: increasing education from 10 to 14 years is associated with a 1.2-point increase in predicted knowledge.

5.4 Another example with a different outcome

- Suppose an article has (Y =) turnout percentage in a district and (X =) campaign spending (in thousands of dollars).
- If the coefficient is (b = 0.8), then:
    - A $10,000 increase in spending ((X = 10)) leads to a predicted turnout change of (0.8 = 8) percentage points.
- Students can always ask: "For a realistic change in (X), how big is the predicted change in (Y)?"

6. Very brief: other models in articles

6.1 Why you might see models that are not OLS

- Some outcomes are not continuous numbers (e.g., yes/no, counts, ordered categories).
- Authors use other models that fit the outcome type better.
- Common ones in political science:

- Logistic regression (logit)
- Probit
- Count models (Poisson, negative binomial)
- Ordered logit/probit

6.2 What stays the same conceptually

- There is still an equation linking (X) to a prediction for (Y).
- The model still has **coefficients** that describe how changes in (X) affect the prediction.
- We can still think in terms of:
  - Coefficient () effect of one-unit change in (X).
  - Bigger change in (X) () coefficient times that change.

6.3 What changes compared to OLS

- In OLS, the line is directly in the units of (Y).
- In logit/probit, the equation predicts something like a *transformed* version of the probability (not raw (Y) itself).
- Authors often help by converting results into:
  - Predicted probabilities for particular values of (X).
  - "Marginal effects" that say "a one-unit change in (X) changes the probability of (Y=1) by so many percentage points."

6.4 How students can read these models at this level

- Focus on the **direction**: is the coefficient positive or negative?
- Focus on whether authors say the effect is "large," "small," or "not statistically significant."
- When authors provide predicted probabilities or marginal effects, read those the same way you did for OLS:
  - For a change in (X), how much does the predicted probability change?

7. Wrap-up and practice

7.1 Key takeaways

- OLS chooses a single regression line to summarize the relationship between (X) and (Y).
- The line becomes a simple formula: (= a + bX).
- Given the coefficient (b), a change in (X) of size (X) leads to a predicted change in (Y) of (b X).

- Other models in articles still link (X) to a prediction; you can read direction and size of effects even without the math.

7.2 Practice Table

Variable	Coefficient	Std. Error
Political interest	1.50	0.30
Constant	20.00	2.00

Y political knowledge index on a 0–10 scale, X political interest on a 1–5 scale.

7.3 Practice questions

1. Write the regression line formula based on the table.
2. Write the effect in a sentence: “A one-unit increase in political interest is associated with a ___-point increase in the predicted political knowledge index.”
3. What is the predicted Y, political knowledge index, for someone with a political interest score of 3?
4. If political interest increases from 2 to 4, what is the predicted change in the political knowledge index?